

AN EXPLICIT FIXED-DIVISOR BOUND FOR A211420

ABSTRACT. Let

$$a_n = \frac{(8n)! n!}{(4n)!(3n)!(2n)!} \quad \text{and} \quad L_r = \text{lcm}(1, 2, \dots, 8r - 1).$$

For every $n \geq 0$, $r \geq 1$, and $k \in \{1, 2, 3\}$, we prove

$$\prod_{i=1}^r (kn + i) \mid L_r^r a_n.$$

This gives an explicit uniform version of a positive-constant divisibility assertion recorded by Peter Bala in OEIS A211420. The weaker existence of some constant follows from Landau's multivariate factorial-ratio criterion; the contribution here is the explicit L_r^r bound and a direct prime-power proof. The theorem has been formalized in Lean 4 with mathlib and checked by Lean's kernel. The constant is not claimed to be sharp.

1. INTRODUCTION

Consider the factorial ratio

$$a_n = \frac{(8n)! n!}{(4n)!(3n)!(2n)!} \quad (n \geq 0).$$

Peter Bala recorded in OEIS A211420 the assertion that, for fixed $r \geq 1$ and $k \in \{1, 2, 3\}$, some positive integer $C(k, r)$, independent of n , makes $C(k, r)a_n$ divisible by the consecutive product $\prod_{i=1}^r (kn + i)$ for every $n \geq 0$ [6]. We give the following explicit choice.

Theorem 1. *Let $n \geq 0$, $r \geq 1$, and $k \in \{1, 2, 3\}$. Then a_n is an integer and, with*

$$L_r = \text{lcm}(1, 2, \dots, 8r - 1), \quad D_{k,r}(n) = \prod_{i=1}^r (kn + i),$$

one has

$$D_{k,r}(n) \mid L_r^r a_n.$$

In particular, the constant L_r^r is positive and uniform in n .

The bound is not claimed to be minimal or sharp. The integrality of a_n was already known: it is the $a = 4$, $b = 1$ specialization of an infinite family in Bober's classification of integral factorial ratios [2]. The weaker existence of some $C(k, r)$ also follows from the classical multivariate criterion of Landau [1]; a two-variable specialization gives, for example, the much larger choice $((8r)!)^r$. The contribution here is the explicit refinement to L_r^r and a direct proof that tracks the relevant prime powers. Rodriguez-Villegas gives broader background on integral factorial ratios and their hypergeometric series [3].

Two published special cases are worth noting. Sun's result, under the relevant specialization, gives $(3n + 1) \mid 5a_n$ [4]. Yang's theorem with $a = 4$ and $b = 1$ gives $(2n + 1)(2n + 3) \mid 99a_n$ for positive n [5]; its two factors form an odd progression rather than the consecutive product in Theorem 1. These comparisons are contextual only. We make no claim of historical priority for Theorem 1.

2. PROOF OF THE EXPLICIT BOUND

Fix $n \geq 0$, $r \geq 1$, and $k \in \{1, 2, 3\}$, and retain the notation of Theorem 1.

For a prime p , we initially interpret v_p as the usual valuation on positive rational numbers. Define

$$\Delta(x) = \lfloor 8x \rfloor + \lfloor x \rfloor - \lfloor 4x \rfloor - \lfloor 3x \rfloor - \lfloor 2x \rfloor.$$

Date: July 2026.

Since $8 + 1 - 4 - 3 - 2 = 0$, the identity $\lfloor d(x+1) \rfloor = \lfloor dx \rfloor + d$ for integral d shows that Δ is one-periodic.

Lemma 2. *The function Δ takes only the values 0 and 1, and its support modulo one is exactly*

$$\left[\frac{1}{8}, \frac{1}{3}\right) \cup \left[\frac{3}{8}, \frac{1}{2}\right) \cup \left[\frac{5}{8}, \frac{2}{3}\right) \cup \left[\frac{7}{8}, 1\right).$$

Moreover, for every prime p and $n \geq 0$,

$$v_p(a_n) = \sum_{j \geq 1} \Delta\left(\frac{n}{p^j}\right),$$

the sum is finite, and a_n is an integer.

Proof. On $[0, 1)$, direct evaluation between the breakpoints of the five floor functions gives the displayed support and shows that the only values are 0 and 1. In particular, $\Delta(x) \geq 0$ for all x .

Legendre's formula, applied to each factorial in a_n , gives the stated valuation identity. The sum is finite: if $p^j > 8n$, then every argument of a floor occurring in the corresponding Legendre summand is smaller than 1, so the summand vanishes. Thus every prime valuation of the positive rational number a_n is nonnegative. Equivalently, the denominator of the factorial ratio divides its numerator, and hence $a_n \in \mathbb{N}$. $\square$

For an integer $q \geq 1$, put

$$N_q = \#\{i \in \{1, \dots, r\} : q \mid kn + i\}.$$

Lemma 3. *For every prime p ,*

$$v_p(D_{k,r}(n)) = \sum_{j \geq 1} N_{p^j}.$$

If $q \geq 8r$, then $N_q \leq \Delta(n/q)$. If $q < 8r$, then $N_q \leq r$. Finally,

$$v_p(L_r) = \#\{j \geq 1 : p^j < 8r\}.$$

Proof. Each factor $kn + i$ contributes one to N_{p^j} for every $1 \leq j \leq v_p(kn + i)$. Summing first over factors or first over exponents therefore gives the first identity; this is precisely why repeated prime powers are counted with their full multiplicity.

Suppose $q \geq 8r$. Since $q > r$, two distinct terms among $kn + 1, \dots, kn + r$ cannot both be divisible by q , so $N_q \leq 1$. If $q \mid kn + i$, write $kn + i = mq$ and set $\varepsilon = i/(kq)$. Then $n/q = m/k - \varepsilon$, where $0 < \varepsilon \leq 1/(8k)$. According to k and the residue class of m , the fractional part of n/q lies in one of the following intervals:

k	$m \pmod{k}$	$\{n/q\}$
1	0	$[7/8, 1)$
2	0, 1	$[15/16, 1), [7/16, 1/2)$
3	0, 1, 2	$[23/24, 1), [7/24, 1/3), [5/8, 2/3)$

Every interval is contained in the support from Lemma 2. The left endpoints are included, notably when $q = 8r$ and $i = r$, while the right endpoints are not attained because $i > 0$. Hence a divisible term forces $\Delta(n/q) = 1$, proving $N_q \leq \Delta(n/q)$.

For $q < 8r$, the bound $N_q \leq r$ is immediate. Finally, the highest power of p dividing the least common multiple of $1, \dots, 8r - 1$ is the largest $p^j < 8r$. Counting its possible exponents gives the last identity. $\square$

Proof of Theorem 1. Lemma 2 proves the integrality of a_n before any integer divisibility involving it is invoked. For each prime p , split the identity of Lemma 3 at $8r$ and apply its two bounds:

$$\begin{aligned} v_p(D_{k,r}(n)) &= \sum_{p^j < 8r} N_{p^j} + \sum_{p^j \geq 8r} N_{p^j} \\ &\leq r v_p(L_r) + \sum_{p^j \geq 8r} \Delta(n/p^j) \\ &\leq r v_p(L_r) + \sum_{j \geq 1} \Delta(n/p^j) \\ &= v_p(L_r^r a_n). \end{aligned}$$

The second inequality uses $\Delta \geq 0$ on the omitted small powers. Thus the valuation of $D_{k,r}(n)$ does not exceed that of $L_r^r a_n$ at any prime, and prime-by-prime comparison proves the desired divisibility. Since $r \geq 1$, the least common multiple L_r is a positive integer, and therefore $L_r^r > 0$. $\square$

3. FORMAL VERIFICATION AND PROVENANCE

The primary Lean development first proves the division-free theorem `A211420.strengthened_cross`. Exact factorial-ratio divisibility is established by `A211420.integral_cross` before the natural-number quotient is introduced, and the quotient identity is then used to derive the final theorem `A211420.strengthened`. The repository pins both Lean and mathlib. Running `./scripts/verify.sh` rebuilds the development and rejects forbidden proof escapes in project Lean files; Lean’s kernel checks the formal proof.

A separate second machine-generated Lean formalization is preserved in the repository. It is not described as independently prompted or as human review.

The ordinary-language proof of the explicit L_r^r strengthening was generated with OpenAI GPT-5.6 Pro. OpenAI Codex produced the primary Lean 4 formalization and repository infrastructure; Lean’s kernel checked the development in the pinned environment. Danny initiated, curated, validated, and published the project. No claim of historical priority or human peer review is made.

REFERENCES

- [1] E. Landau, *Sur les conditions de divisibilité d’un produit de factorielles par un autre*, Nouvelles annales de mathématiques, 3e série, **19** (1900), 344–362.
- [2] J. W. Bober, *Factorial ratios, hypergeometric series, and a family of step functions*, J. London Math. Soc. (2) **79** (2009), 422–444.
- [3] F. Rodriguez-Villegas, *Integral ratios of factorials and algebraic hypergeometric functions*, arXiv:math/0701362 (2007).
- [4] Z.-W. Sun, *On divisibility of binomial coefficients*, J. Aust. Math. Soc. **93** (2012), 189–201.
- [5] Q.-H. Yang, *Proof of a conjecture of Amdeberhan and Moll on a divisibility property of binomial coefficients*, Electron. J. Combin. **22** (2015), Paper P1.9.
- [6] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, Sequence A211420, <https://oeis.org/A211420>.